

MINI OPERATIONS ERP

Full-Stack Developer Technical Case Study

Operations workflow

Inventory → Work Order → Stock Check → Internal Transfer → Customer Reservation

A functional operations management system focused on inventory correctness, role-based access, work-order material checks, stock transfers, and customer stock reservation.

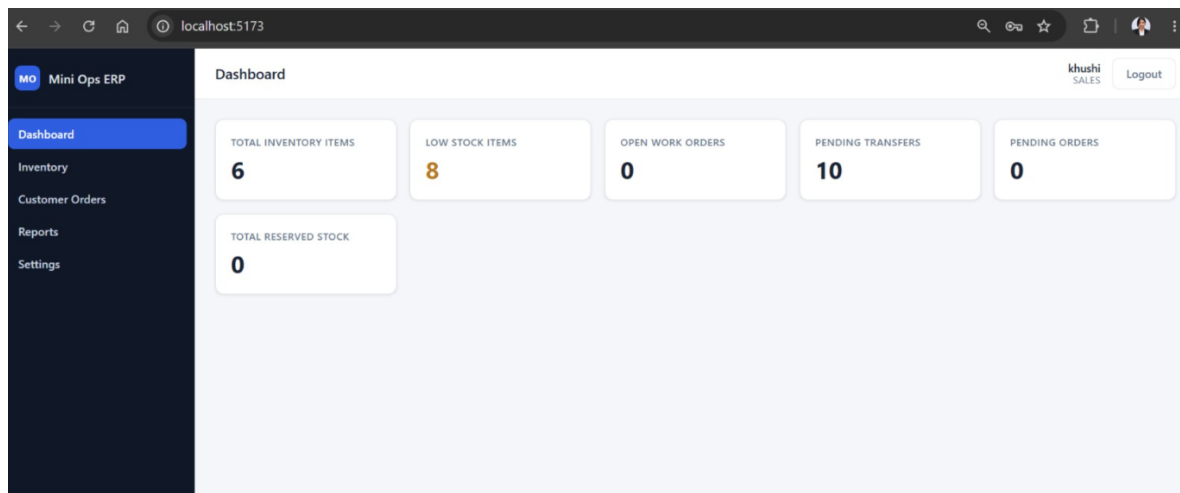


Figure 1. Operations dashboard overview

1. Executive Summary

Mini Operations ERP is designed around the business flow defined in the technical case study: inventory is tracked by item, category, location and batch; work orders can expose material shortages; available stock can be moved through internal transfers; and customer orders reserve stock without allowing reservations beyond availability.

The submitted interface demonstrates the core operational screens and provides visible validation feedback for insufficient stock conditions. Transfer records visibly distinguish REQUESTED and RECEIVED states, while the inventory view separates physical, reserved and available quantities.

Area	Case-study requirement	Visible implementation evidence
Authentication	Login + role-based access	Sign-in screen and employee role listing
Inventory	Physical / reserved / available stock	Inventory table with quantity columns
Work Orders	Required, available, shortage, status	Work Orders table
Transfers	Requested → Dispatched → Received	Transfer list and status controls
Reservations	Prevent reservation beyond availability	Insufficient-stock validation message
Reporting	Operational visibility	Dashboard and low-stock report

2. Authentication & Role-Based Access

The application begins with a dedicated sign-in interface. The employee management screen also exposes distinct roles including ADMIN, OPERATIONS and SALES, aligning with the case-study requirement for backend authorization and minimum role separation.

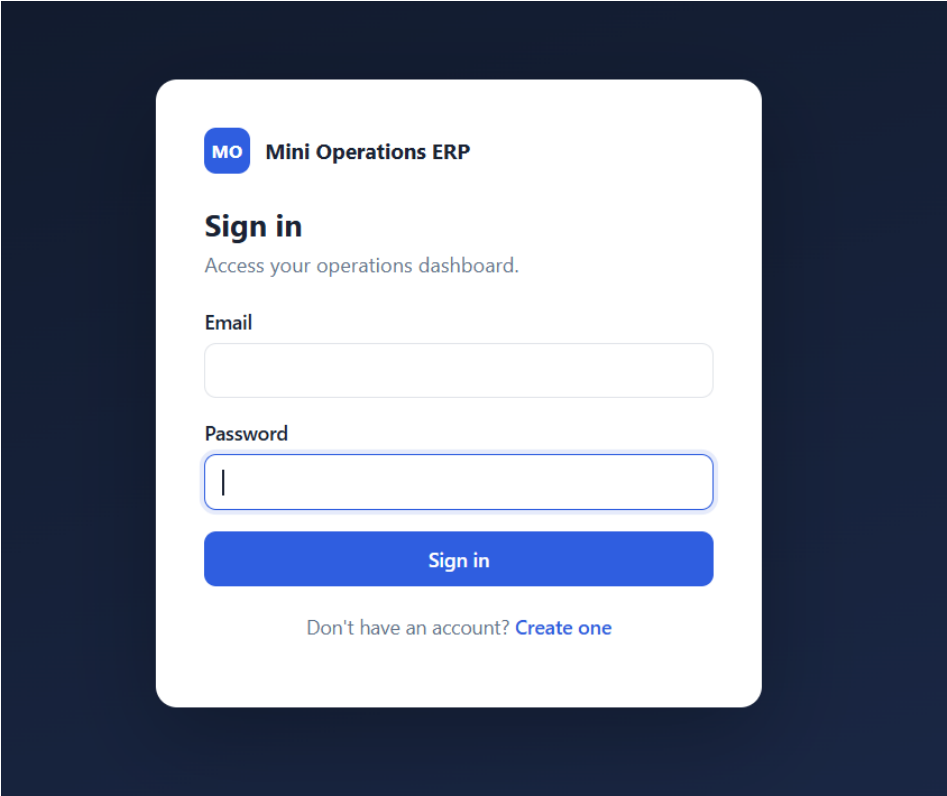


Figure 2. User sign-in interface

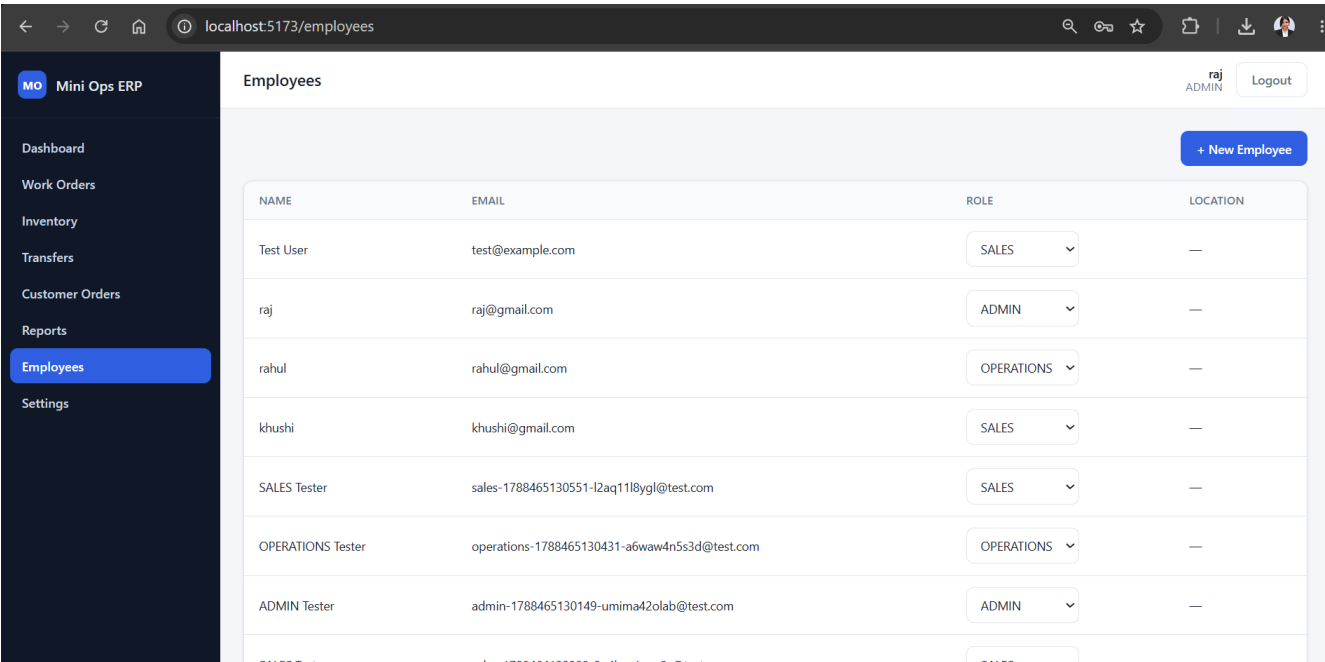


Figure 3. Employee and role management

3. Inventory Management

The Inventory module presents item-level stock records with category, location and batch information. Physical, reserved and available quantities are shown independently, making the stock position auditable and allowing availability to be used as the operational quantity for reservations and transfers.

Mini Ops ERP

Dashboard

Inventory

Customer Orders

Reports

Settings

Inventory

khushi SALES Logout

All locations

All categories

Search item name...

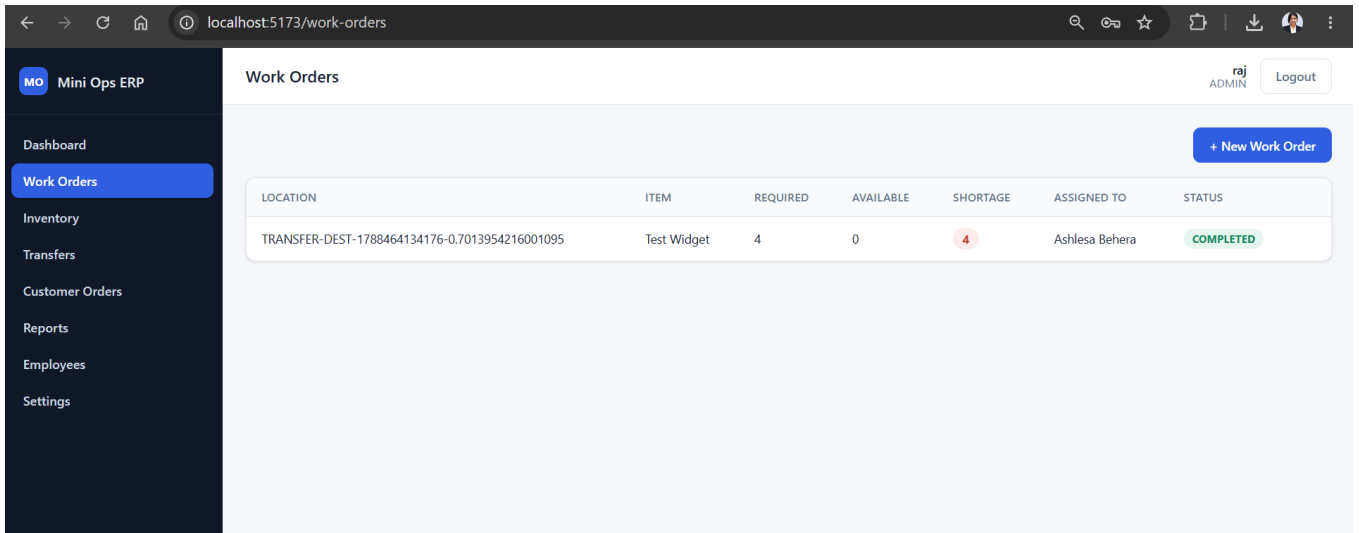
ITEM	CATEGORY	LOCATION	BATCH	PHYSICAL	RESERVED	AVAILABLE
Test Widget	Cat-1788465130651	WH-1788465130658-A	B1-1788465130664	0	0	0
Transfer Widget-1788465130876	TransferCat-1788465130871-0.4032140779911213	TRANSFER-SOURCE-1788465130879-0.019457051416151838	B3-1788465130886	30	0	30
Test Widget	Cat-1788464133980	WH-1788464133984-A	B1-1788464133988	0	0	0
Transfer Widget-1788464134169	TransferCat-1788464134166-0.4710142435841649	TRANSFER-SOURCE-1788464134173-0.5804296683957526	B3-1788464134178	30	0	30
Test Widget	Cat-1788463965007	WH-1788463965011-A	B1-1788463965015	0	0	0
Test Widget	Cat-1788463808670	WH-1788463808680-A	B1-1788463808686	0	0	0
Test Widget	Cat-1788465130651	WH-1788465130660-B	TRANSFER-295fb5aa	10	0	10
Test Widget	Cat-1788465130651	WH-1788465130658-A	B4-1788465130966	50	0	50
Transfer Widget-	TransferCat-1788465130871-	TRANSFER-DEST-1788465130883-	TRANSFER-			

Figure 4. Inventory records with physical, reserved and available quantities

Operational controls demonstrated: location/category filtering, item search, batch-level records, and visible available-stock indicators.

4. Work Orders & Material Stock Check

The Work Orders screen surfaces the fields needed to evaluate material requirements: location, item, required quantity, available quantity, shortage, assigned user and status. In the captured example, a required quantity of 4 is compared with 0 available, producing a shortage of 4 while the work order is marked COMPLETED.



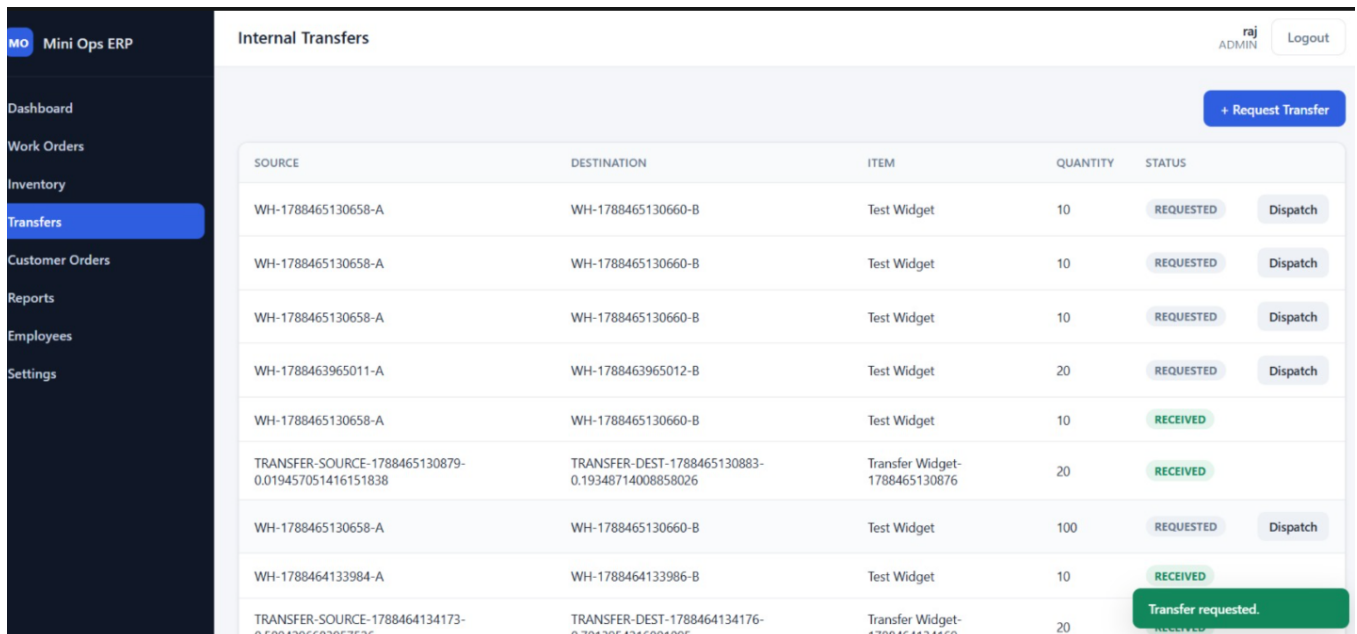
The screenshot shows a web application interface for 'Mini Ops ERP'. The left sidebar contains navigation links: Dashboard, Work Orders (highlighted), Inventory, Transfers, Customer Orders, Reports, Employees, and Settings. The main content area is titled 'Work Orders' and includes a '+ New Work Order' button. Below this is a table with the following columns: LOCATION, ITEM, REQUIRED, AVAILABLE, SHORTAGE, ASSIGNED TO, and STATUS. A single row of data is displayed, showing a location ID, the item 'Test Widget', a required quantity of 4, an available quantity of 0, a shortage of 4 (highlighted in red), assigned to 'Ashlesha Behera', and a status of 'COMPLETED' (highlighted in green).

LOCATION	ITEM	REQUIRED	AVAILABLE	SHORTAGE	ASSIGNED TO	STATUS
TRANSFER-DEST-1788464134176-0.7013954216001095	Test Widget	4	0	4	Ashlesha Behera	COMPLETED

Figure 5. Work Order material availability and shortage visibility

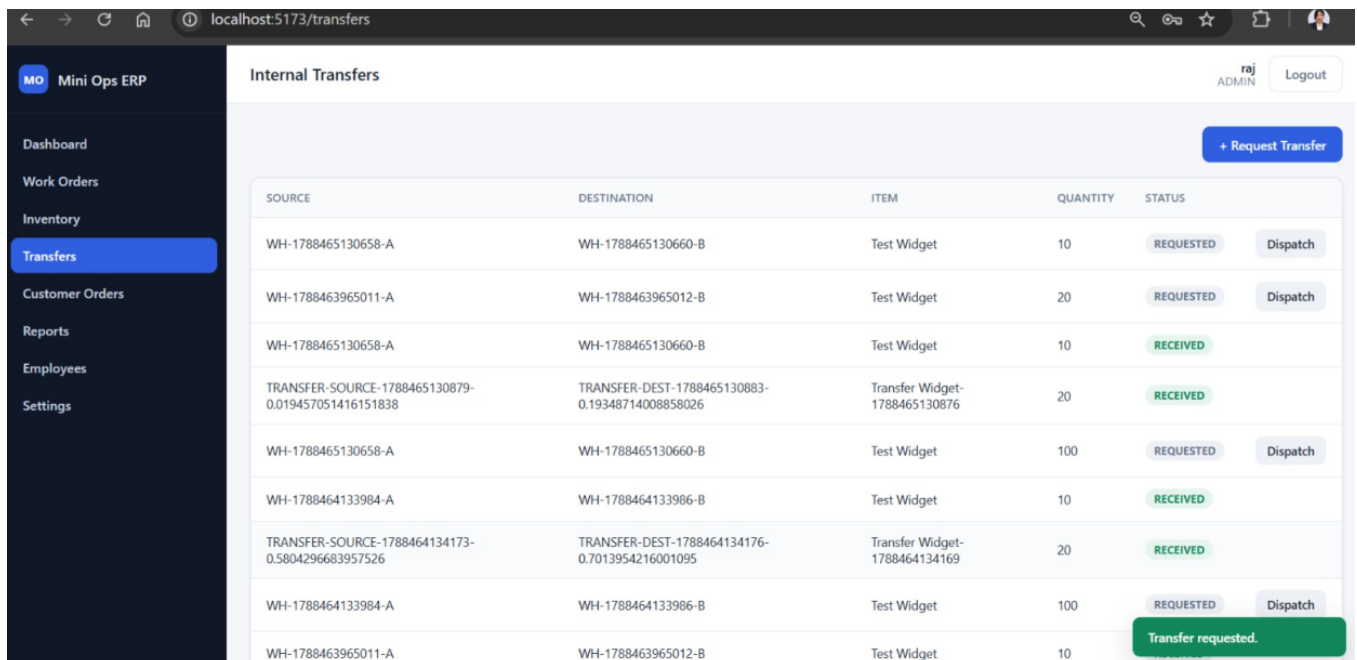
5. Internal Stock Transfer

Internal Transfers model movement between a source and destination location. The captured screens show requested transfers and completed receipts, with status explicitly represented as REQUESTED or RECEIVED. This supports the required lifecycle in which source stock is affected at dispatch and destination stock is increased only upon receipt.



SOURCE	DESTINATION	ITEM	QUANTITY	STATUS	
WH-1788465130658-A	WH-1788465130660-B	Test Widget	10	REQUESTED	Dispatch
WH-1788465130658-A	WH-1788465130660-B	Test Widget	10	REQUESTED	Dispatch
WH-1788465130658-A	WH-1788465130660-B	Test Widget	10	REQUESTED	Dispatch
WH-1788463965011-A	WH-1788463965012-B	Test Widget	20	REQUESTED	Dispatch
WH-1788465130658-A	WH-1788465130660-B	Test Widget	10	RECEIVED	
TRANSFER-SOURCE-1788465130879-0.019457051416151838	TRANSFER-DEST-1788465130883-0.19348714008858026	Transfer Widget-1788465130876	20	RECEIVED	
WH-1788465130658-A	WH-1788465130660-B	Test Widget	100	REQUESTED	Dispatch
WH-1788464133984-A	WH-1788464133986-B	Test Widget	10	RECEIVED	
TRANSFER-SOURCE-1788464134173-0.5804296683957526	TRANSFER-DEST-1788464134176-0.7013954216001095	Transfer Widget-1788464134169	20	RECEIVED	

Figure 6. Internal Transfer queue with requested and received states

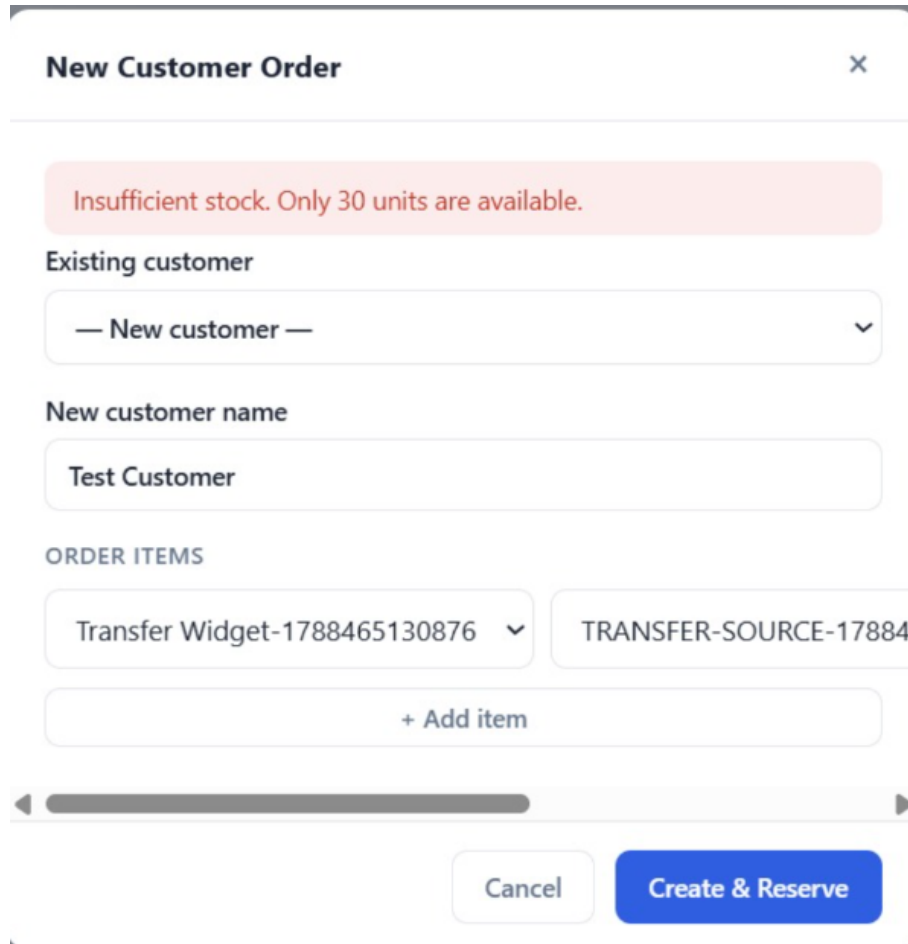


SOURCE	DESTINATION	ITEM	QUANTITY	STATUS	
WH-1788465130658-A	WH-1788465130660-B	Test Widget	10	REQUESTED	Dispatch
WH-1788463965011-A	WH-1788463965012-B	Test Widget	20	REQUESTED	Dispatch
WH-1788465130658-A	WH-1788465130660-B	Test Widget	10	RECEIVED	
TRANSFER-SOURCE-1788465130879-0.019457051416151838	TRANSFER-DEST-1788465130883-0.19348714008858026	Transfer Widget-1788465130876	20	RECEIVED	
WH-1788465130658-A	WH-1788465130660-B	Test Widget	100	REQUESTED	Dispatch
WH-1788464133984-A	WH-1788464133986-B	Test Widget	10	RECEIVED	
TRANSFER-SOURCE-1788464134173-0.5804296683957526	TRANSFER-DEST-1788464134176-0.7013954216001095	Transfer Widget-1788464134169	20	RECEIVED	
WH-1788464133984-A	WH-1788464133986-B	Test Widget	100	REQUESTED	Dispatch
WH-1788463965011-A	WH-1788463965012-B	Test Widget	10	REQUESTED	Dispatch

Figure 7. Transfer records and dispatch controls

6. Customer Orders & Stock Reservation

The customer-order workflow includes customer selection, order items and a Create & Reserve action. The captured validation state explicitly blocks an order when the requested quantity exceeds available stock, showing the available quantity to the user.



The screenshot shows a web form titled "New Customer Order" with a close button (X) in the top right corner. A red error message at the top states: "Insufficient stock. Only 30 units are available." Below this, the form has two sections: "Existing customer" with a dropdown menu currently showing "— New customer —" and a downward arrow, and "New customer name" with a text input field containing "Test Customer". The "ORDER ITEMS" section contains two input fields: the first is a dropdown menu showing "Transfer Widget-1788465130876" with a downward arrow, and the second is a text input field containing "TRANSFER-SOURCE-17884". Below these is a button labeled "+ Add item". At the bottom of the form are two buttons: "Cancel" and "Create & Reserve".

Figure 8. Customer Order reservation blocked by insufficient stock

Observed validation: the interface reports that only 30 units are available when the attempted reservation exceeds that quantity. This is direct UI evidence of the required reservation guard.

7. Operational Reporting & Dashboard

The dashboard provides a concise operational snapshot, including total inventory items, low-stock items, open work orders, pending transfers, pending orders and total reserved stock. The Reports screen expands this into a low-stock report with item, location, batch and available quantity.

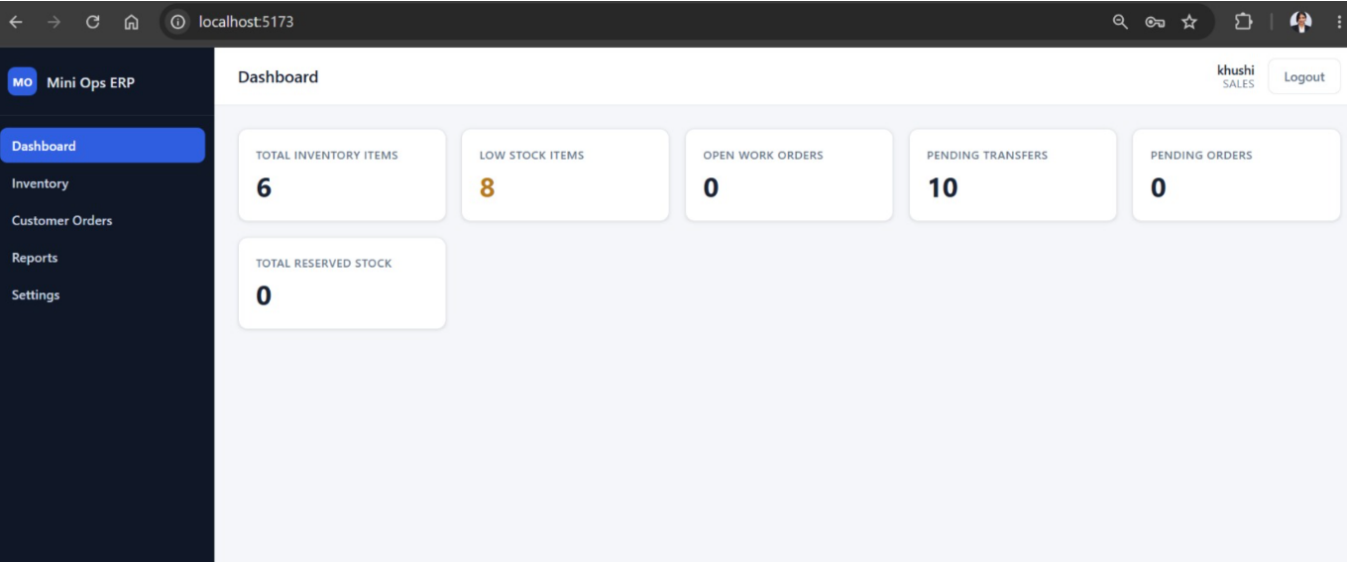


Figure 9. Operations dashboard KPIs

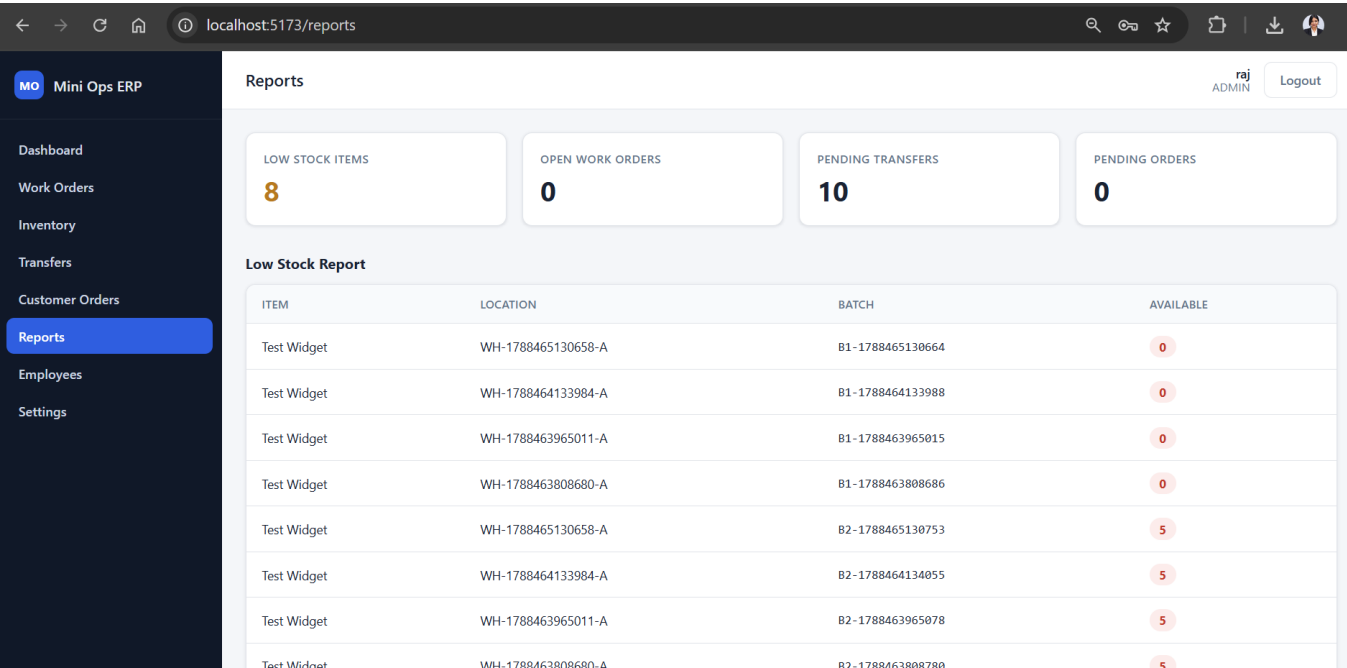


Figure 10. Low-stock operational report

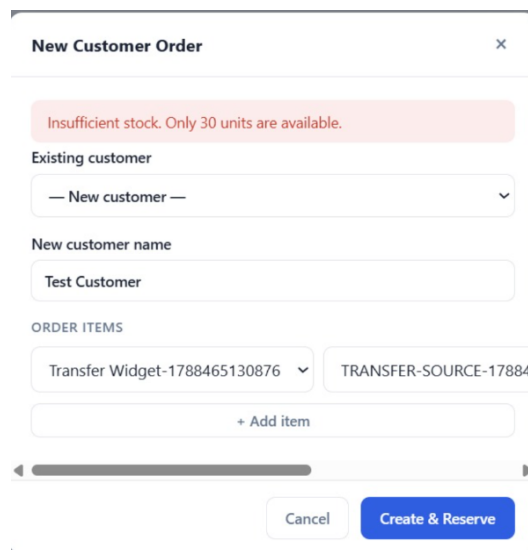
8. Mandatory Test Cases & Evidence

The case study mandates five minimum tests covering inventory reservation, transfers, receipt timing, duplicate receipt protection and authorization. The screenshots below are organized as evidence of the corresponding validation and state-management behavior.

Test 1 — Cannot reserve more than available inventory

Expected: reservation must fail when requested quantity exceeds available stock.

Evidence: The customer-order form displays an explicit insufficient-stock message and the available quantity.



The screenshot shows a web form titled "New Customer Order" with a close button (X) in the top right corner. A red error message at the top states: "Insufficient stock. Only 30 units are available." Below this, the form has sections for "Existing customer" (a dropdown menu currently showing "— New customer —"), "New customer name" (a text input field containing "Test Customer"), and "ORDER ITEMS" (a section with a dropdown menu showing "Transfer Widget-1788465130876" and a text input field containing "TRANSFER-SOURCE-17884", followed by a "+ Add item" button). At the bottom of the form are two buttons: "Cancel" and "Create & Reserve".

Figure 11. Reservation validation evidence

Test 2 — Cannot transfer more than available inventory

Expected: dispatch must fail when the source location lacks sufficient available stock.

Evidence: The transfer screen displays an explicit source-stock validation error.

MO Mini Ops ERP

Internal Transfers

raj ADMIN Logout

+ Request Transfer

SOURCE	DESTINATION	ITEM	QUANTITY	STATUS	
WH-1788463965011-A	WH-1788463965012-B	Test Widget	20	REQUESTED	Dispatch
WH-1788465130658-A	WH-1788465130660-B	Test Widget	10	RECEIVED	
TRANSFER-SOURCE-1788465130879-0.019457051416151838	TRANSFER-DEST-1788465130883-0.19348714008858026	Transfer Widget-1788465130876	20	RECEIVED	
WH-1788465130658-A	WH-1788465130660-B	Test Widget	100	REQUESTED	Dispatch
WH-1788464133984-A	WH-1788464133986-B	Test Widget	10	RECEIVED	
TRANSFER-SOURCE-1788464134173-0.5804296683957526	TRANSFER-DEST-1788464134176-0.7013954216001095	Transfer Widget-1788464134169	20	RECEIVED	
WH-1788464133984-A	WH-1788464133986-B	Test Widget	100	REQUESTED	Dispatch
WH-1788463965011-A	WH-1788463965012-B	Test Widget	10	RECEIVED	
WH-1788463965011-A	WH-1788463965012-B	Test Widget			Insufficient available stock at source location to dispatch this transfer.

Figure 12. Transfer stock validation evidence

Test 3 — Destination stock increases only after transfer receipt

Expected: destination quantity changes on receipt, not merely on request.

Evidence: Transfer records visibly distinguish REQUESTED and RECEIVED states, supporting the receipt-based lifecycle.

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Mini Ops ERP

Dashboard

Work Orders

Inventory

Transfers

Customer Orders

Reports

Employees

Settings

Internal Transfers

rajADMINLogout

+ Request Transfer

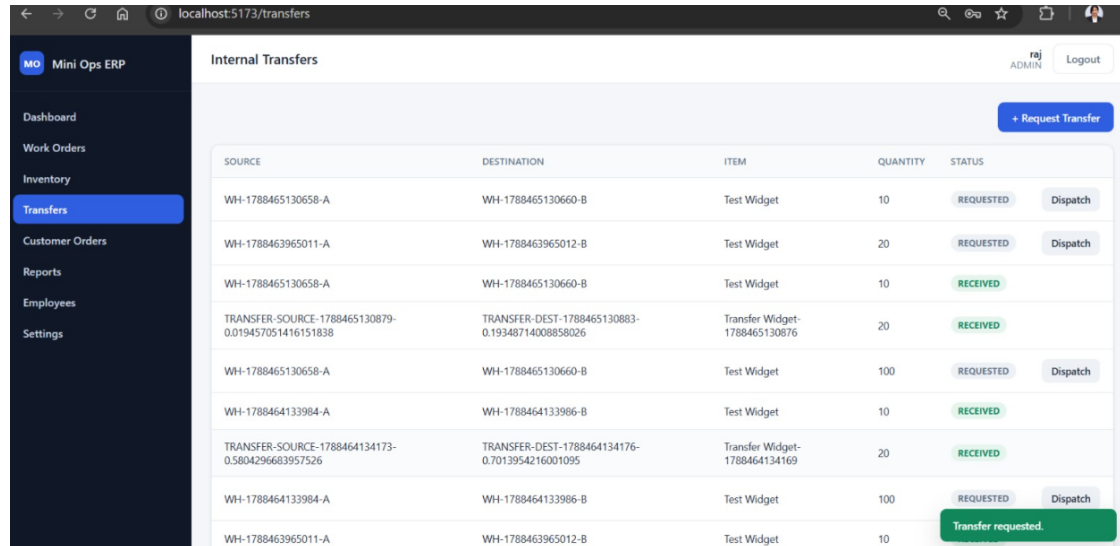
SOURCE	DESTINATION	ITEM	QUANTITY	STATUS	
WH-1788465130658-A	WH-1788465130660-B	Test Widget	10	REQUESTED	Dispatch
WH-1788465130658-A	WH-1788465130660-B	Test Widget	10	REQUESTED	Dispatch
WH-1788465130658-A	WH-1788465130660-B	Test Widget	10	REQUESTED	Dispatch
WH-1788463965011-A	WH-1788463965012-B	Test Widget	20	REQUESTED	Dispatch
WH-1788465130658-A	WH-1788465130660-B	Test Widget	10	RECEIVED	
TRANSFER-SOURCE-1788465130879-0.019457051416151838	TRANSFER-DEST-1788465130883-0.19348714008858026	Transfer Widget-1788465130876	20	RECEIVED	
WH-1788465130658-A	WH-1788465130660-B	Test Widget	100	REQUESTED	Dispatch
WH-1788464133984-A	WH-1788464133986-B	Test Widget	10	RECEIVED	
TRANSFER-SOURCE-1788464134173-0.5804296683957526	TRANSFER-DEST-1788464134176-0.7013954216001095	Transfer Widget-1788464134169	20	REQUESTED	Transfer requested.

Figure 13. Transfer lifecycle evidence

Test 4 — Same transfer cannot be received twice

Expected: a transfer already in RECEIVED state cannot be received again.

Evidence: Received transfers remain represented as RECEIVED rather than presenting the dispatch action used by requested transfers.



Internal Transfers

raj ADMIN Logout

+ Request Transfer

SOURCE	DESTINATION	ITEM	QUANTITY	STATUS	
WH-1788465130658-A	WH-1788465130660-B	Test Widget	10	REQUESTED	Dispatch
WH-1788463965011-A	WH-1788463965012-B	Test Widget	20	REQUESTED	Dispatch
WH-1788465130658-A	WH-1788465130660-B	Test Widget	10	RECEIVED	
TRANSFER-SOURCE-1788465130879-0.019457051416151838	TRANSFER-DEST-1788465130883-0.19348714008858026	Transfer Widget-1788465130876	20	RECEIVED	
WH-1788465130658-A	WH-1788465130660-B	Test Widget	100	REQUESTED	Dispatch
WH-1788464133984-A	WH-1788464133986-B	Test Widget	10	RECEIVED	
TRANSFER-SOURCE-1788464134173-0.5804296603957526	TRANSFER-DEST-1788464134176-0.7013954216001095	Transfer Widget-1788464134169	20	RECEIVED	
WH-1788464133984-A	WH-1788464133986-B	Test Widget	100	REQUESTED	Dispatch
WH-1788463965011-A	WH-1788463965012-B	Test Widget	10		

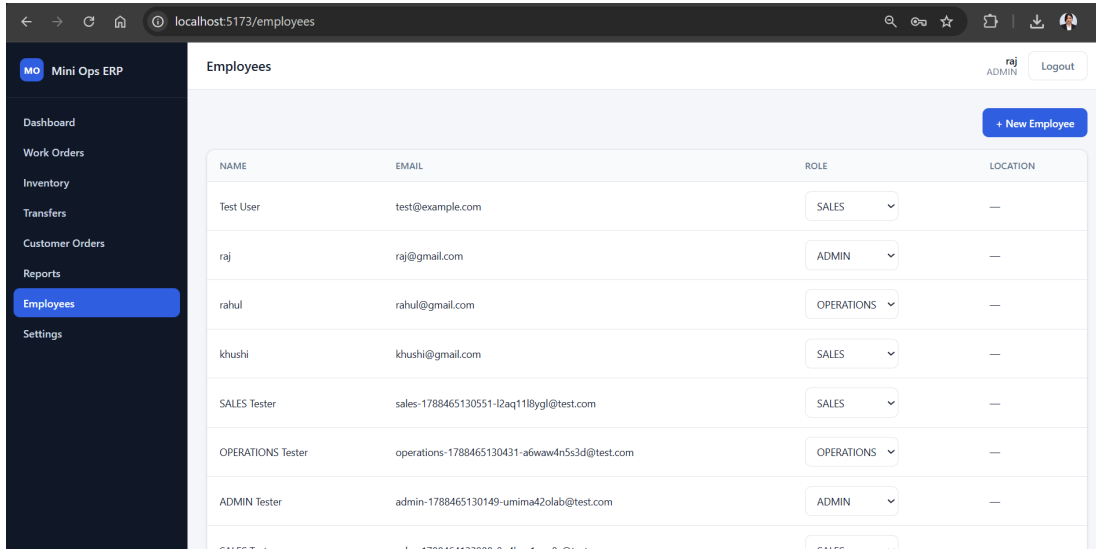
Transfer requested.

Figure 14. Received-state evidence

Test 5 — Unauthorized user cannot perform restricted operation

Expected: role-based authorization must restrict protected backend operations.

Evidence: The employee screen demonstrates explicit ADMIN, OPERATIONS and SALES roles. The screenshot itself does not show a failed unauthorized API action, so the authorization result should be verified from the test/API logs as well.



The screenshot shows a web application interface for 'Mini Ops ERP'. The left sidebar contains a menu with items: Dashboard, Work Orders, Inventory, Transfers, Customer Orders, Reports, Employees (highlighted), and Settings. The main content area is titled 'Employees' and features a '+ New Employee' button. Below this is a table with columns: NAME, EMAIL, ROLE, and LOCATION. The table lists several users with roles including SALES, ADMIN, and OPERATIONS.

NAME	EMAIL	ROLE	LOCATION
Test User	test@example.com	SALES	—
raj	raj@gmail.com	ADMIN	—
rahul	rahul@gmail.com	OPERATIONS	—
khushi	khushi@gmail.com	SALES	—
SALES Tester	sales-1788465130551-l2aq118ygl@test.com	SALES	—
OPERATIONS Tester	operations-1788465130431-afwaw4n5s3d@test.com	OPERATIONS	—
ADMIN Tester	admin-1788465130149-umima42olab@test.com	ADMIN	—

Figure 15. Role separation evidence

9. Technical Alignment & Conclusion

The submitted implementation visibly covers the principal functional surfaces requested in the case study: login, inventory, work orders, internal transfers, customer orders, employee roles, dashboard metrics and operational reporting. The captured validation states also demonstrate attention to business-rule enforcement rather than UI-only behavior.

The case study places particular weight on backend/API correctness, database design, business logic and inventory/transaction correctness. These areas should therefore be supported in the final repository and API documentation with the corresponding implementation details, transaction boundaries and automated test output.

Evaluation area	Weight
Backend & APIs	20
Database Design	15
Business Logic	20
Inventory / Transaction Correctness	15
Frontend Integration	10
Authentication & Authorization	8
Testing	5
Code Quality	5
Documentation	2
Total	100

Prepared as a professional project/case-study document from the supplied case-study specification and application screenshots.